

# AWS EBS Volume – Overview and How to Increase Size

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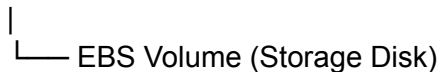
## 1 What is an EBS Volume?

Amazon EBS (Elastic Block Store) is a **block storage service** in AWS used with EC2 instances.

- It works like a **virtual hard disk** attached to a server.
- It stores:
  - Operating System files
  - Application data
  - Logs
  - Databases

### Example Structure:

EC2 Instance



## 2 Why Increase EBS Volume Size?

Sometimes disk space becomes full.

### Common reasons:

- Application logs increase over time
- Database data grows
- Docker images and containers use space
- Application files and uploads increase

### Example:

Total Disk : 20GB

Used Disk : 19GB

Free Space : 1GB

### 3 Troubleshooting (Before Increasing EBS)

Before increasing the volume, check where the disk space is used.

#### Check Disk Space:

```
df -h
```

Shows:

- Total size
- Used space
- Available space

#### Example:

| Filesystem     | Size | Used | Avail | Use% |
|----------------|------|------|-------|------|
| /dev/nvme0n1p1 | 20G  | 19G  | 1G    | 95%  |

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#### Check Folder Size:

```
du -sh /*
```

#### Example:

```
2.1G /var
5.3G /home
7.2G /opt
```

#### Check Logs:

```
du -sh /var/log
```

👉 Delete unnecessary files if possible.

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### 4 Steps to Increase EBS Volume Size

#### Step 1 – Open EC2

- Login to AWS Console
  - Go to EC2 Dashboard
  - Click **Instances**
  - Select your instance
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## Step 2 – Open Storage

- Scroll down
- Click **Storage tab**
- Find attached **Volume ID**
- Click on Volume ID

Example:

Volume ID: vol-xxxxxxx

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## Step 3 – Modify Volume

- Click **Modify**
- Enter new size

Example:

Old Size : 20 GB

New Size : 40 GB

- Click **Modify** and confirm

👉 AWS will start resizing:

modifying → optimizing → completed

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## 5 Extend File System on Server

After increasing size in AWS, update the filesystem in Linux.

### Step 1 – Connect to Server

```
ssh -i key.pem ubuntu@server-ip
```

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## Step 2 – Check Disk Info

lsblk  
df -h

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## Step 3 – Extend Partition

`sudo growpart /dev/nvme0n1 1`

This command expands (increases) the size of a partition after you increase disk size in AWS.

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## Step 4 – Resize Filesystem

`sudo resize2fs /dev/nvme0n1p1`

this command resizes (expands or shrinks) the filesystem on the partition `/dev/nvme0n1p1` so it can use the full space of the disk.

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## Step 5 – Verify

df -h

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## 6 Important Commands

### lsblk

- command is used to list all block storage devices like disks and partitions in a tree format.

### df -h

- is used to check **disk space usage** in the system.

### du -sh

- is used to show total size of a directory in human-readable format.

## Disk Example (lsblk Output)

```
nvme0n1    8G disk
└─nvme0n1p1 7G part /
nvme1n1    8G disk /data
```

### Explanation:

| Name      | Description                  |
|-----------|------------------------------|
| nvme0n1   | First disk (root EBS volume) |
| nvme0n1p1 | Partition of disk            |
| nvme1n1   | Second attached EBS volume   |

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## Final Summary

- EBS is a **virtual disk in AWS**
- Increase size when disk is full
- Always check usage before increasing
- After resizing in AWS, update filesystem in Linux

### Process:

Check → Increase Volume → Extend Partition → Resize Filesystem → Verify

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